

Flow Past a Circular Cylinder: Reynolds Number Sweep

OpenFOAM v2012 · pimpleFoam / pimpleFoam + kOmegaSST · 4-Core Parallel · snappyHexMesh

A parametric study of 2D flow over a circular cylinder (diameter = 1 m) across a sweep of Reynolds numbers, from creeping Stokes flow through laminar vortex shedding onset into the turbulent regime. Developed as part of an OpenFOAM CFD portfolio demonstrating transient flow simulation and the diagnostic process behind getting a solver to reproduce known physics.

Key Results

Case	Re	Inlet U (m/s)	Model	Cd	Cl amplitude	Regime
Re_3	3	4.5e-05	laminar	5.92	~0	Steady, unseparated creeping flow
Re_40	40	6.0e-04	laminar	1.64	~0	Steady, fixed symmetric recirculation bubble
Re_100	100	1.5e-03	laminar	1.35	0.31	Periodic von Karman vortex shedding
Re_30000	3.0e4	0.45	kOmegaSST	1.21	1.09	Turbulent subcritical shedding
Re_800000	8.0e5	12	kOmegaSST	1.27	1.22	Supercritical turbulent shedding

Cd is the final drag coefficient, Cl amplitude is the peak-to-peak lift coefficient swing divided by two, taken over the last 20% of each run.

Domain and mesh

All cases share the same geometry and mesh recipe:

- Domain: a $[-10, 20] \times [-10, 10] \times [0, 1]$ m box, cylinder of radius 0.5 m centred at the origin.
- `blockMesh` generates a uniform background hex mesh.
- `snappyHexMesh` cuts the cylinder out of the background mesh using a built-in `searchableCylinder` primitive (no STL needed), with surface and distance-based refinement near the wall and in the wake.

- inlet/outlet are patch, front/back are empty (2D case), cylinder is wall. top/bottom are patch with a slip field condition, not symmetryPlane (see lessons below).
- Around 115,000 cells per case.

snappyHexMesh is run in parallel across 4 cores (decomposePar then mpirun -np 4 snappyHexMesh -overwrite -parallel, followed by reconstructParMesh), which cuts meshing time from roughly 30 minutes to under two minutes per case versus a serial run, with identical mesh quality.

This mesh approach was chosen after an earlier hand-built O-grid blockMeshDict (multiple blocks stitched directly with explicit vertices) produced a topologically disconnected mesh, checkMesh reported three separate, unconnected cell regions from non-conformal seams between the inner O-grid blocks and the far-field blocks. Rebuilding around snappyHexMesh removed that whole class of bug.

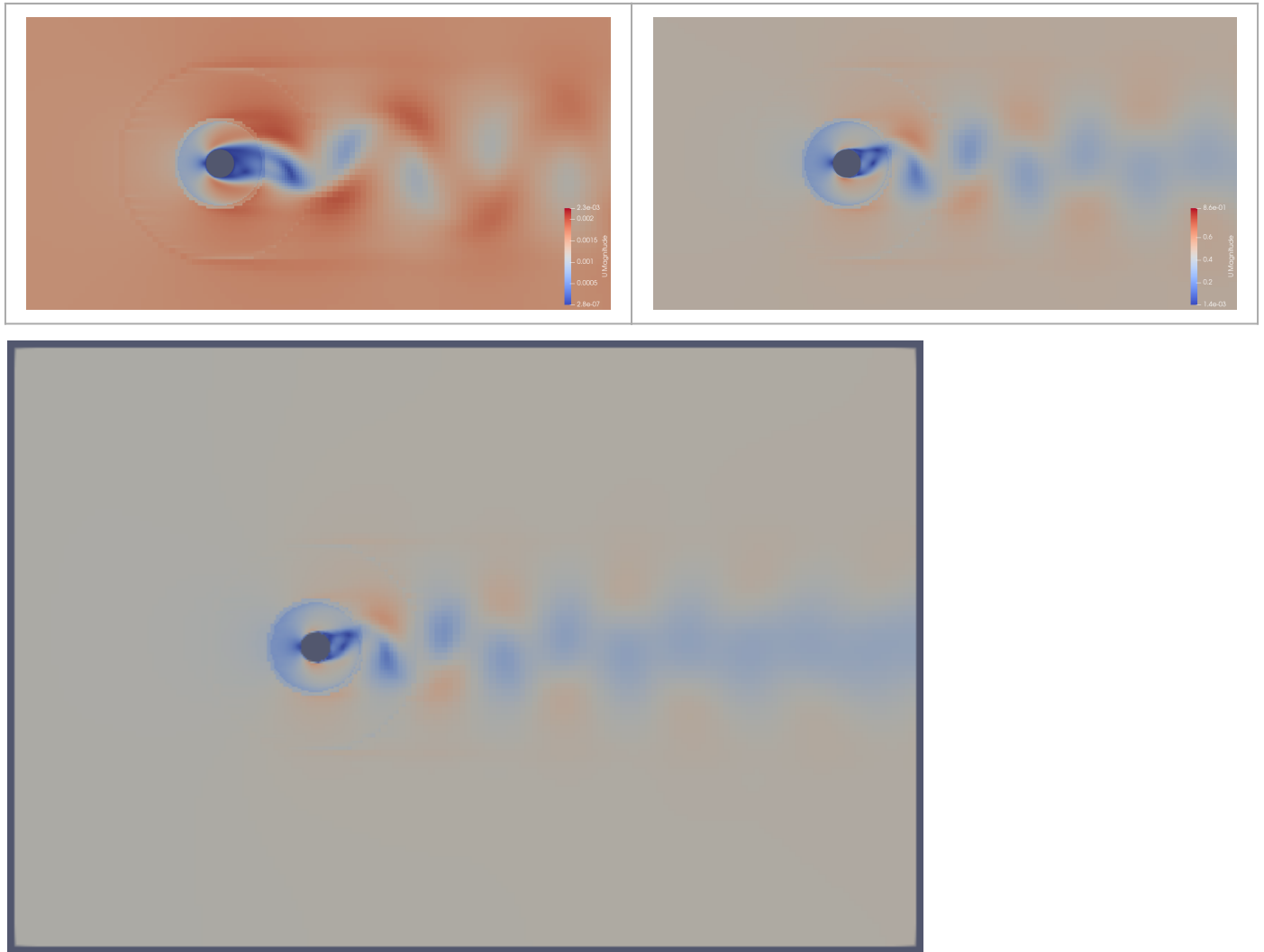
Solver setup

- **Solver:** pimpleFoam, transient PIMPLE algorithm, no under-relaxation.
 - **Turbulence:** laminar for Re_3/Re_40/Re_100, kOmegaSST RAS model for the higher-Re cases.
 - **Viscosity:** fixed at $\nu = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$ for every case, with inlet velocity scaled to hit each target Reynolds number through $\text{Re} = U \cdot D / \nu$.
 - **Force monitoring:** forceCoeffs function object on the cylinder patch, sampled every 10 timesteps.
 - **Time step:** adjustable, Courant-limited ($\text{maxCo} \ 0.8$), with a per-case maxDeltaT ceiling tuned to the case's own convective timescale.
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Results

Velocity contours

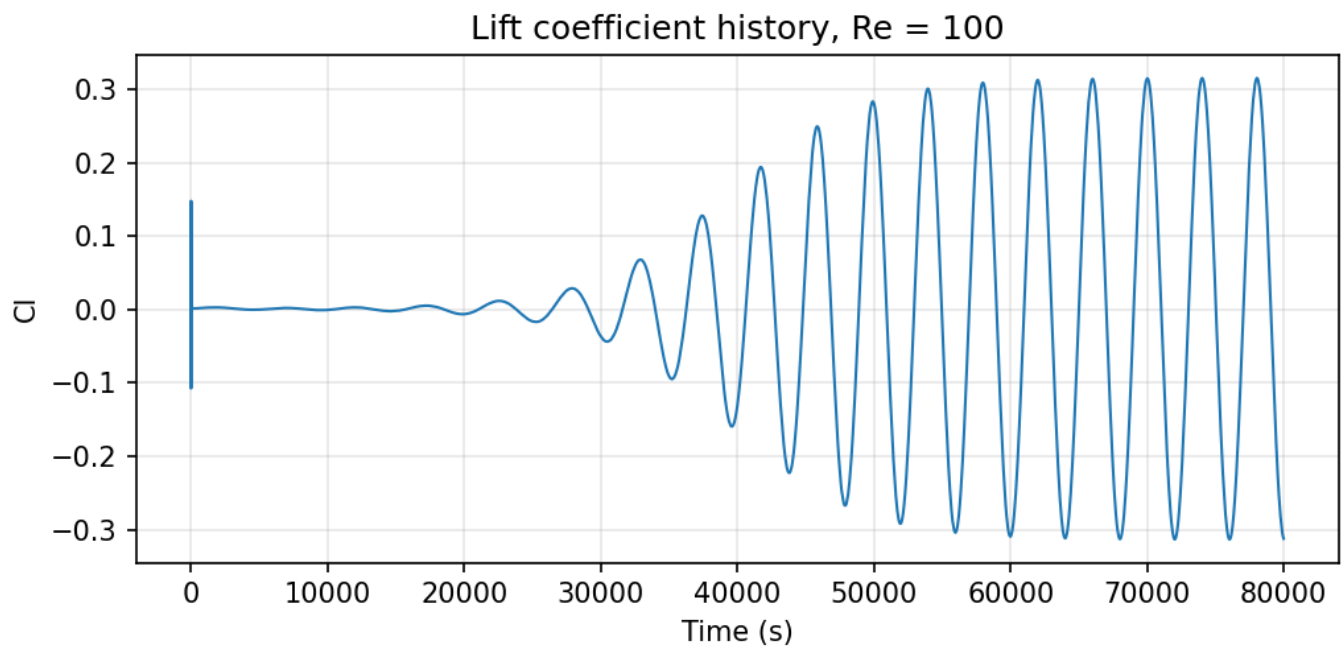




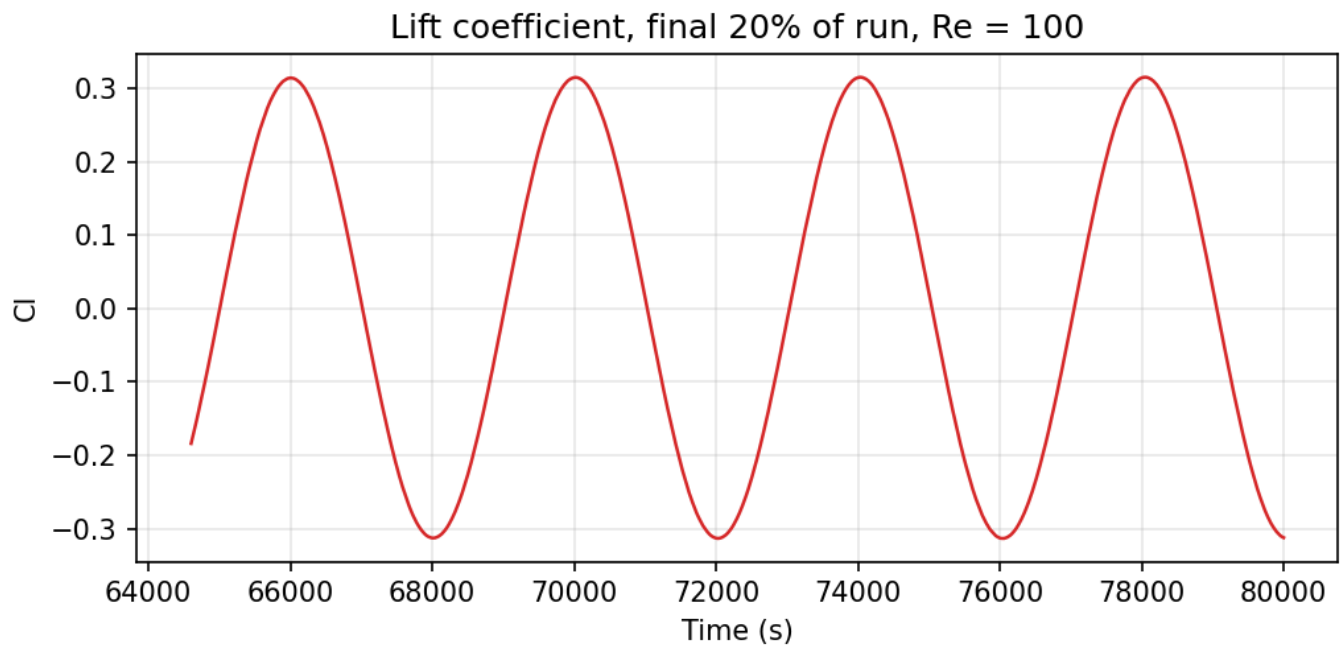
$Re = 800,000$ velocity magnitude, showing a compact turbulent wake with tighter vortex spacing relative to the lower- Re cases.

At $Re = 3$ and $Re = 40$ the wake stays attached and symmetric about the centreline, matching the unseparated and fixed-bubble regimes expected at these Reynolds numbers. At $Re = 100$ and above, the wake breaks symmetry and sheds alternating vortices, the signature von Karman street. At $Re = 800,000$ the wake becomes more compact and turbulent, consistent with the supercritical regime.

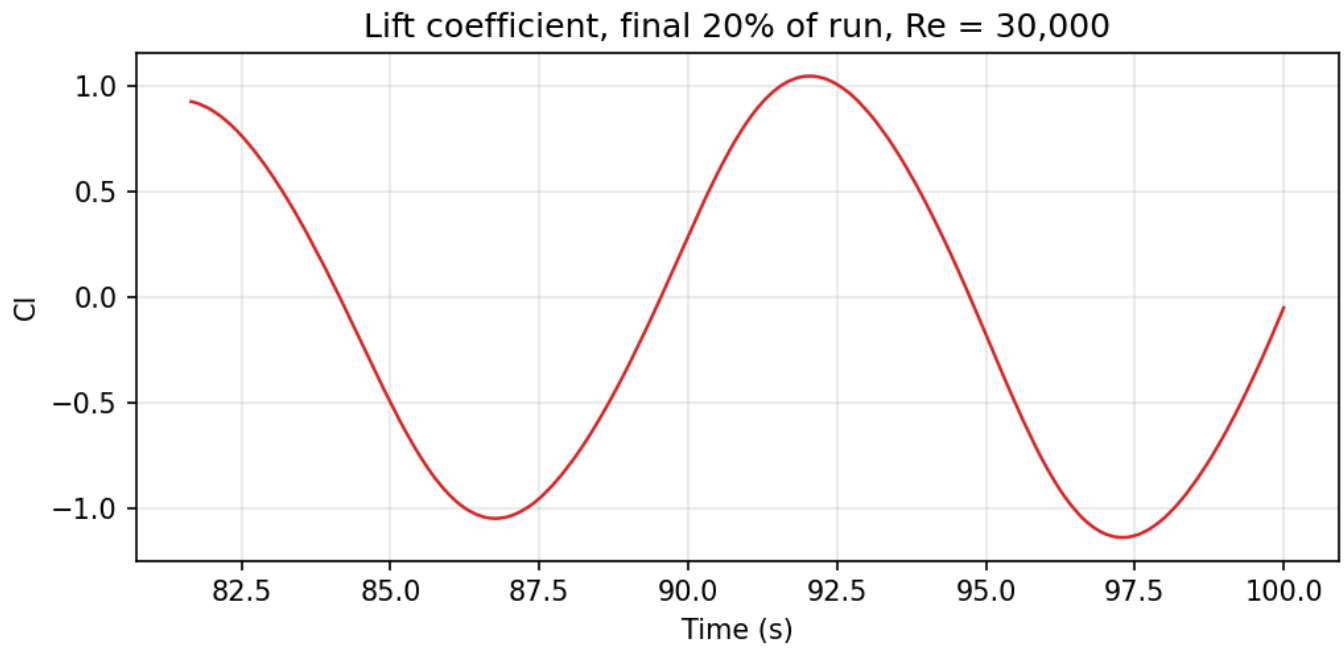
Lift coefficient history



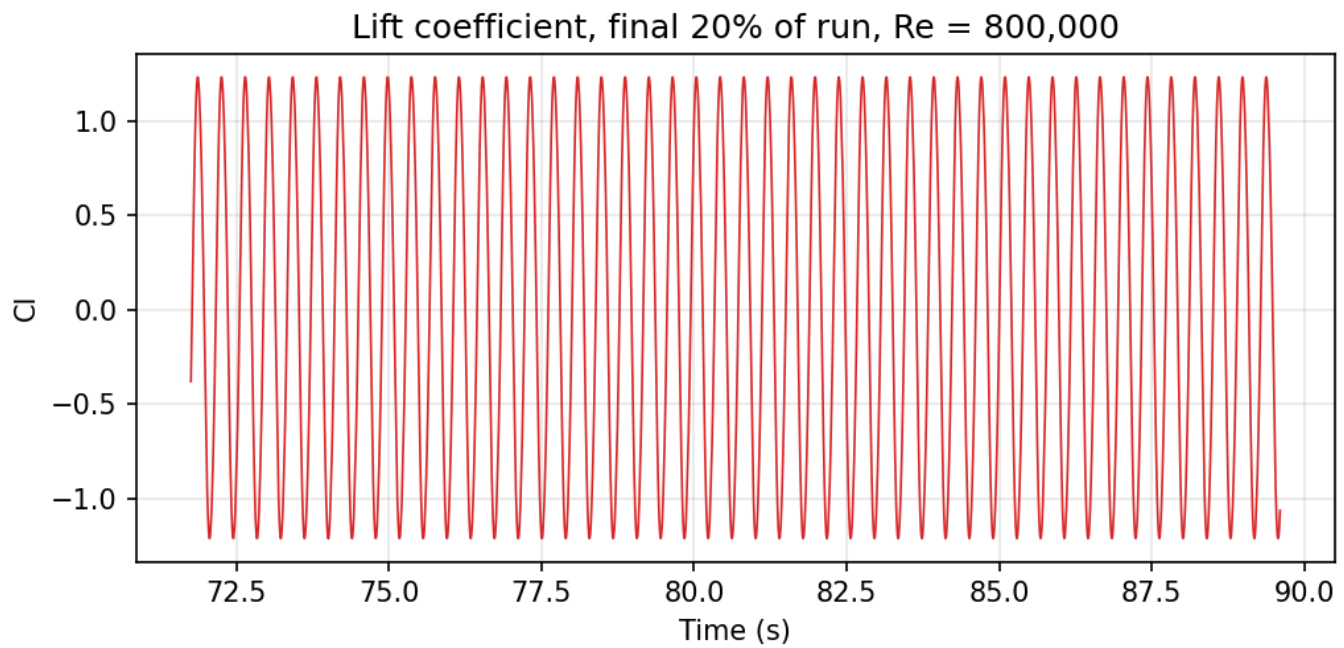
Re = 100 lift coefficient over the full run. C_l starts near zero, grows as the shedding instability develops, then settles into a clean periodic oscillation.



Final 20% of the Re = 100 run, showing the fully developed shedding cycle, period around 4000 s, amplitude about 0.31.

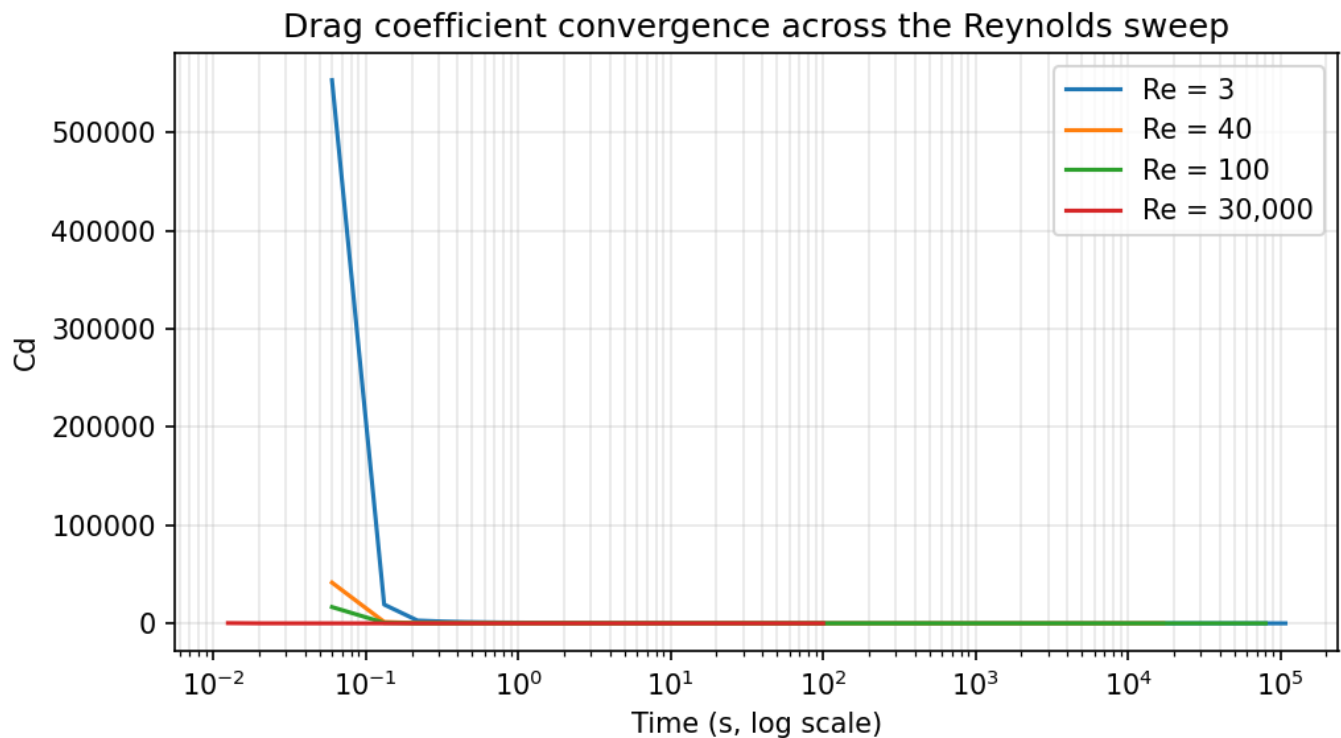


$Re = 30,000$ lift coefficient over the last 20% of its run. Shedding is faster and stronger in this turbulent case, the amplitude is still growing toward saturation at the current end time.



$Re = 800,000$ fully saturated periodic shedding over the last 20% of the run, amplitude around 1.22, with a noticeably higher frequency than $Re = 30,000$ due to the much higher inlet velocity.

Drag coefficient convergence



Drag coefficient versus time (log scale) for all four completed cases. Re_3 converges the slowest, a known feature of low-Reynolds-number cylinder drag, while the higher-Re cases settle within their much shorter convective timescales.

Workflow

```
blockMesh (background hex mesh) | decomposePar (background mesh only) | mpirun -np 4 snappyHexMesh
-overwrite -parallel | reconstructParMesh -constant -mergeTol 1e-6 | checkMesh | rm -rf processor* &&
decomposePar (re-decompose 0/ fields against final mesh) | mpirun -np 4 pimpleFoam -parallel |
reconstructPar -> ParaView post-processing
```

The second `decomposePar` pass is required because the cylinder patch only exists in the mesh after parallel `snappyHexMesh` finishes, decomposing the fields before that point leaves the per-processor field files without a `cylinder` boundary entry, which crashes `pimpleFoam` with a `Cannot find patchField entry for cylinder` error.

Debugging lessons

This case went through three rounds of fixes before it produced believable physics, all three were necessary, none alone was sufficient:

`symmetryPlane` top/bottom boundaries silently forced a symmetric solution. The original mesh and field boundary conditions used `symmetryPlane` on the top and bottom domain walls. That condition mathematically enforces a mirror-symmetric flow field, which makes the asymmetric von Karman shedding mode impossible no matter how long the case runs. Fixed by changing the patch type to `patch` in `blockMeshDict` and the field condition to `slip` in `0/U`, `0/p`, `0/k`, `0/omega`, `0/nut`.

Leftover steady-state relaxation factors damped the transient solution. `fvSolution` carried a `relaxationFactors` block (`p 0.3, U/k/omega 0.7`) left over from an earlier steady-state attempt. Applying SIMPLE-style under-relaxation to a transient PIMPLE run artificially damps the time-accurate update each step and suppresses real unsteady dynamics. Removed entirely.

endTime was far shorter than the flow's own timescale. Inlet velocity is scaled down to hit low Reynolds numbers, which makes the convective timescale D/U and shedding period $\tau \sim D/(St \cdot U)$ very large in absolute seconds. The original `endTime = 200 s` was less than 1/20th of one shedding cycle at $Re = 100$. Increased `endTime` (and the `maxDeltaT` ceiling, to keep wall-clock cost down) per case based on its own convective and shedding timescale.

A coarse maxDeltaT produced timestep-locked numerical ringing, not real oscillation. After the fixes above, $Re = 3$ still showed an oscillating `CI`, but at constant amplitude with a period of exactly five timesteps, regardless of the (well within bounds) Courant number. That is temporal under-resolution producing numerical ringing around the true steady solution, not physical shedding, which should not exist at $Re = 3$. Lowering `maxDeltaT` from 50 s to 5 s removed the ringing and let `CI` settle to its expected near-zero steady value.

The takeaway: a clean-looking, bounded oscillation in a force coefficient history is not on its own proof of physical vortex shedding, it can just as easily be a numerical artifact, and only checking it against the expected flow regime (steady vs unsteady, oscillation period vs timestep) catches the difference.

Running a case yourself

Each `Re_*` folder is a self-contained OpenFOAM case with everything needed to regenerate the mesh and resolve the flow from scratch. Only the generated mesh (`constant/polyMesh`), the parallel decomposition (`processor*`), and intermediate time directories have been omitted to keep the repo lightweight, the final time directory is kept as a sample result.

```
bash cd "Re_100" ./Allrun
```

Turbulent cases (`Re_300000`, `Re_800000`, `Re_5000000`) use `kOmegaSST` and require a `wallDist` entry in `system/fvSchemes` (included), a common gotcha when adapting `fvSchemes` from a laminar case.

Limitations

- `Re_800000` was stopped at $t = 89.6$ s (out of a planned 100 s end time) once the `CI` history confirmed fully saturated periodic shedding, so the final `Cd/CI` values are representative of the converged solution.
 - 2D domain with `empty` front/back patches, no spanwise three-dimensional shedding effects are captured.
 - Fixed kinematic viscosity with inlet velocity scaled per case, rather than holding velocity fixed and varying viscosity, this keeps the geometry and mesh identical across the sweep but means each case has a very different physical timescale.
 - $Re = 3$ drag coefficient is still drifting by a fraction of a percent per 2000 s of simulated time at the point captured here, a known feature of how slowly low-Reynolds-number cylinder drag approaches its asymptote (related to the Oseen/Stokes paradox).
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Software Stack

Tool	Version	Purpose
OpenFOAM	v2012	CFD solver
ParaView	5.11	Post-processing and contour rendering
Python / matplotlib	3.x	Force coefficient plots
Ubuntu (WSL2)	24.04	OS
OpenMPI	4.0.3	Parallel communication

Repository

Part of the [OpenFOAM Portfolio](#), a collection of CFD simulations demonstrating thermal and fluid simulation skills relevant to thermal engineering roles.